

Chapter 2.8 | Linear Approximation ; Differentials

recall t.l.e. $y - f(a) = f'(a)(x-a)$

This can be viewed as an approximation to the curve $y=f(x)$ near $x=a$

then $f(x) - f(a) = f'(a)(x-a)$ when x is close to a

or $f(x) \approx f(a) + f'(a)(x-a)$

this is called the linear approximation of f at a or tangent line approximation.

Ex | find the linearization $L(x)$, of the function $f(x) = x^5 + 4x^3$ at $a = -1$

$$L(x) = f(a) + f'(a)(x-a)$$

$$= -5 + 12(x - (-1))$$

$$= -5 + 12(x+1)$$

$$= -5 + 12x + 12$$

$$= 12x + 7$$

$$f(-1) = (-1)^5 + 4(-1)^3$$

$$= -1 - 4$$

$$= -5$$

$$f'(x) = 5x^4 + 12x^2$$

$$f'(-1) = 5(-1)^4 + 12(-1)^2$$

$$= 5 + 12$$

$$= 17$$

Note: Basically finding tangent line

Ex | Use linear approximation to estimate the given #
 $(1.999)^5$

let $f(x) = x^5$; $a = 2$

Find the linearization

$$f(2) = 2^5 = 32$$

$$f'(x) = 5x^4$$

$$L(x) = f(x) + f'(x)(x-a)$$

$$f'(2) = 80$$

$$= 32 + 80(x-2)$$

$$L(1.999) = 32 + 80(1.999-2)$$

$$= 32 + 80(-.001)$$

$$= 32 - .08$$

$$\approx 31.92$$

Differentials: differential represents an infinitely small change in some quantity

let $y = f(x)$

then the differentiable dx , represents a small change in the x -direction

The differentiable dy , is defined in terms of dx by the eq, $dy = f'(x)dx$

note if $dx \neq 0$ then we get $\frac{dy}{dx} = f'(x)$

Ex | Find the differential of each function

$$y = \tan \sqrt{8t}$$

$$f'(t) = \sec^2(\sqrt{8t}) \cdot (\sqrt{8t})'$$

$$f(t) = \tan \sqrt{8t}$$

$$= \sec^2(\sqrt{8t}) \cdot \frac{1}{2} \cdot (8t)^{-\frac{1}{2}} \cdot (8t)'$$

Double Chain Rule

$$= \sec^2(\sqrt{8t}) \cdot \frac{1}{2} (8t)^{-\frac{1}{2}} \cdot 8$$

$$= \frac{8 \sec^2(\sqrt{8t})}{2 \sqrt{8t}}$$

$$dy = \frac{8 \sec^2(\sqrt{8t})}{2 \sqrt{8t}} dt$$